

# Smart Bus Tracking and Monitoring System Using Camera

## Literature Review

Group No. 6

Akash V G

Fidha Fathima A

Mohammed Ramshad T S

Nithya Manoj

Under the guidance of Rakhi R

# Paper 1: Bus Monitoring System - Neha Parkara et al., IJSART 2021

- **Concept:** IoT–GPS based bus monitoring and tracking
- **Methodology:** Real time tracking using IOT-GPS based system
- **Advantages:** Real-time tracking, improved safety, reduced waiting time
- **Limitations and Gap identified:** Internet/GPS dependency, low security, no identification system

## Paper 2: Attendance System Using Facial Recognition - Tushar Rane et al., IJCRT 2023

- **Concept:** Facial recognition–based automated attendance
- **Methodology:** Attendance is recorded automatically
- **Advantages:** Contactless, fast, prevents proxy
- **Limitations and Gap identified:** Lighting, angle, occlusion issues, no tracking

# Paper 3: AI-Based Bus Attendance System - Johnson C et al., IJNRD 2023

- **Concept:** AI-based bus attendance using facial recognition
- **Methodology:** Attendance is marked using face recognition and barcode scanning
- **Advantages:** Contactless, automated, prevents proxy
- **Limitations and Gap identified:** No real time alerts

# Paper 4: Real-Time Bus Tracking and Attendance - R. Aruna et al., IJSRED 2025

- **Concept:** IoT-based bus tracking with RFID attendance
- **Methodology:** Automated entry and exit are monitored
- **Advantages:** Automated attendance, real-time tracking
- **Limitations and Gap identified:** No computer vision or face recognition

# Paper 5: Smart College Bus Tracker - M. Sowndharya et al., IJNRD 2025

- **Concept:** Real-time college bus tracking with ETA prediction
- **Methodology:** ETA is calculated dynamically
- **Advantages:** Reduced waiting time, improved transparency
- **Limitations and Gap identified:** No attendance

# Summary

| Title                                                                                            | Author                                                             | Methodology                                                                             | Limitation                                                                                            |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Bus Attendance System through Facial Recognition by Implementing AI Integration, IJNRD, Nov 2022 | Johnson C., Karthikeyan S., Sakthivel R., Shankar K., Sakthi A.    | Facial recognition, base64 encoding, student database, attendance marking, notification | Energy-efficient hardware, biometric integration issues, AI integration with student performance data |
| Attendance System Using Facial Recognition (IJERT), May 2023                                     | Tushar Rane, Sagar Dhoot, Nayan Jain, Ramesh Naphade, Rohit Virani | Image capture, attendance marking, email notification                                   | No automated notification system                                                                      |
| Bus Monitoring System (IJERT), April 2021                                                        | Neha Parkara, Ashwini Patil et al.                                 | Bus tracking using smart connectivity                                                   | Additional hardware required, privacy concerns                                                        |
| Smart College Bus Tracking: Real-Time Location & ETA Prediction, IJNRD, 2025                     | M. Soundharya et al.                                               | Live location tracking, ETA prediction                                                  | GPS accuracy and internet dependency                                                                  |
| Real-Time College Bus Tracking and Student Attendance System, IJNRD, 2025                        | R. Anjana et al.                                                   | GPS tracking, automated attendance, alerts                                              | Lighting issues, privacy concerns                                                                     |

# Proposed System

- Automatic attendance using facial recognition
- Live tracking of college buses
- Centralized storage of attendance and location data
- Easy monitoring for administrators
- Reduced manual work and errors



# References

- ① Parkara, N., Patil, A., et al., "Bus Monitoring System," International Journal of Scientific and Academic Research in Technology (IJSART), 2021.
- ② Rane, T., Dhoot, S., Jain, N., Naphade, R., Virani, R., "Attendance System Using Facial Recognition," International Journal of Creative Research Thoughts (IJCRT), 2023.
- ③ Johnson, C., Karthikeyan, S., Sakthivel, R., Shankar, K., Sakthi, A., "Bus Attendance System Through Facial Recognition by Implementing AI Integration," International Journal of Novel Research and Development (IJNRD), 2023.
- ④ Aruna, R., Laksha, M., Prakashini, P., Arun, S., "Real-Time College Bus Tracking and Student Attendance System," International Journal of Scientific Research and Engineering Development (IJSRED), 2025.
- ⑤ Sowndharya, M., Kavya, R., Karthik, S., Ananya, S., "Smart College Bus Tracker: Real-Time Location and ETA Prediction," International Journal of Novel Research and Development (IJNRD), 2025.